DATA SCIENCE PROJECT

EXPLORATORY ANALYSIS / MACHINE LEARNING

Climate Change and Its Relationship with Crop Yield in India

How temperature and rainfall are linked to the yield of Wheat, Rice, Maize and Soybean (2000–2024)

AUTHOR

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TOOLS

Python, Pandas, NumPy, Matplotlib, Scikit-learn, Google Colab

DATA SOURCES

FAOSTAT (crop yield) and NASA POWER (climate)

STUDY SCOPE

India • Four crops • 2000–2024

A DATA-LED LOOK AT CLIMATE PATTERNS AND AGRICULTURAL YIELD

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Summary

Climate change can affect farming. Changes in temperature and rainfall can change how well crops grow.

This project looks at how temperature and rainfall are linked to the yield of four major crops in India: Wheat, Rice, Maize and Soybean. Yield means how much a crop is produced from one hectare of land (kg/ha).

Crop yield data came from FAOSTAT and climate data came from NASA POWER. I used Python (Pandas, NumPy, Matplotlib and Scikit-learn) to clean the data, make graphs and build prediction models.

The data covers 2000 to 2024 and has 100 crop-yield records. I checked the link between climate and yield using correlation, and I built a separate Multiple Linear Regression model for each crop.

The overall model got an R^2 score of 0.984 and an MAE of 124.56 kg/ha. Temperature and rainfall are linked to the yield of each crop in different ways.

Project at a glance

4 CROPS Wheat • Rice • Maize • Soybean	**100** RECORDS 2000–2024
0.984 OVERALL R^2 Combined test-set score	**124.56 kg/ha** OVERALL MAE Average absolute error

1. Introduction

Farming is a big part of India's economy. It gives food and jobs to a large number of people. Crop production depends on many things, such as temperature, rainfall, soil quality, irrigation and farming methods.

Climate change can change temperature and rainfall patterns. Very high heat, uneven rain and drought can make it hard for crops to grow.

In this project, I use data science to study how temperature and rainfall are linked to the yield of four major crops in India. I also use machine learning to build models that predict crop yield from past data.

2. Problem Statement

Changes in climate may affect how much crop we can grow. This makes it harder to plan farming and keep food available.

The goal of this project is to study past temperature, rainfall and crop-yield data and find out how they are related. The project also builds machine learning models that estimate crop yield using climate and year information.

3. Objectives

1. Collect past crop-yield and climate data.
2. Study the yield trends of Wheat, Rice, Maize and Soybean.
3. Study how temperature is related to crop yield.
4. Study how rainfall is related to crop yield.
5. Build machine learning models to predict crop yield.
6. Compare actual yields with predicted yields.
7. Understand how climate changes may affect farming output.

4. Tools Used

Tool	What it was used for
Python	Programming and analysis
Google Colab	Running the Python code
Pandas	Cleaning and handling data
NumPy	Number calculations
Matplotlib	Making graphs
Scikit-learn	Machine learning
FAOSTAT	Source of crop-yield data
NASA POWER	Source of climate data

5. Data Description

I used two datasets in this project.

5.1 Crop Yield Data

Source: FAOSTAT. It has past yield records for the selected crops in India. I kept only the four crops and the years I needed.

Item	Details
Country	India
Crops	Wheat, Rice, Maize and Soybean
Time period	2000–2024
Total records	100
Yield unit	kg/ha

5.2 Climate Data

Source: NASA POWER. It has these variables:

Item	Details
Year	Year of the record
Average Temperature	Yearly average temperature in °C
Total Rainfall	Yearly rainfall in mm
Time period	2000–2024
Yearly records	25

> **Important data note**
> The climate data comes from one central India location. It is a stand-in for the country, not the true average of all of India.

6. Data Cleaning

I followed these steps before starting the analysis:

1. Downloaded the crop data from FAOSTAT.
2. Selected Wheat, Rice, Maize and Soybean.
3. Kept only the records for India from 2000 to 2024.
4. Kept only the columns I needed.
5. Downloaded temperature and rainfall data from NASA POWER.
6. Made yearly climate records.
7. Joined the crop and climate data using the Year column.
8. Checked for missing values and duplicate rows.

6.1 Final Data Check

The final dataset has 100 records, with no missing values and no duplicate rows.

Check	Result
Total records	100
Missing values	0
Duplicate rows	0
Number of crops	4
Years covered	2000–2024

7. Exploring the Data

I explored the data with graphs to understand its patterns.

7.1 Crop Yield Trends

First, I plotted the yield of each crop from 2000 to 2024 to see how it changed.

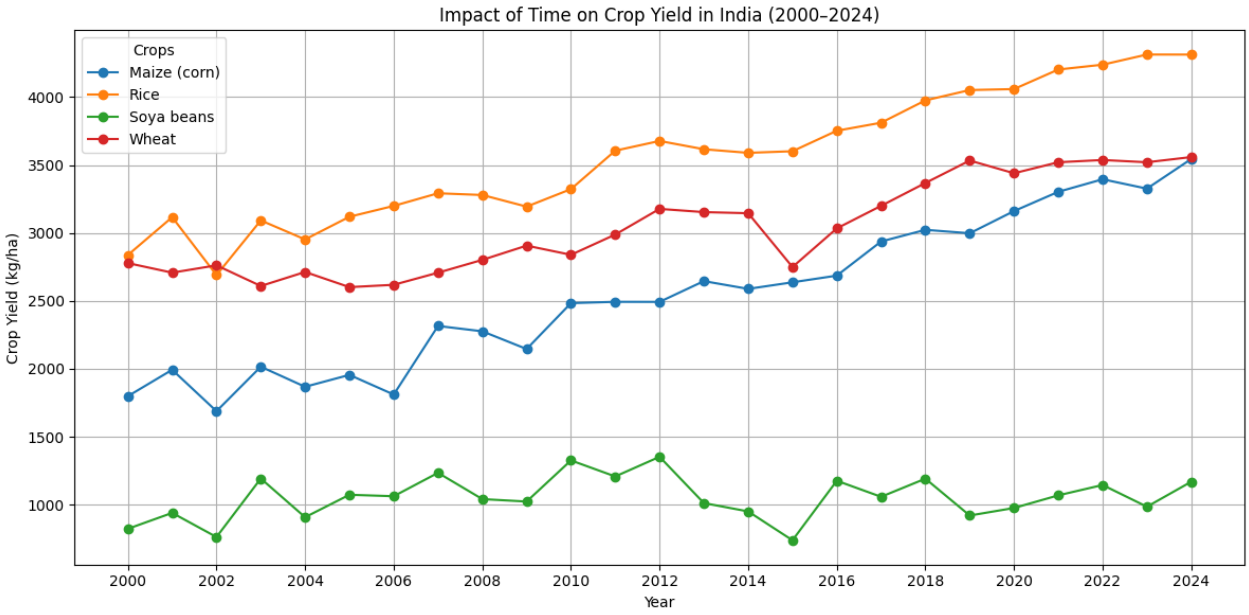


Figure 1: Yearly yield of Maize, Rice, Soybean and Wheat in India (2000–2024)

What the graph shows

- Rice keeps rising, from about 2,800 kg/ha in 2000 to above 4,300 kg/ha in 2024.
- Wheat also rises overall, with ups and downs and a clear dip around 2015.
- Maize goes up from about 1,800 kg/ha to about 3,500 kg/ha, with some year-to-year changes.
- Soybean has the lowest yield and goes up and down without a strong upward trend.

7.2 Temperature and Rainfall vs Yield

Next, I made scatter plots to see how temperature and rainfall relate to the yield of each crop.

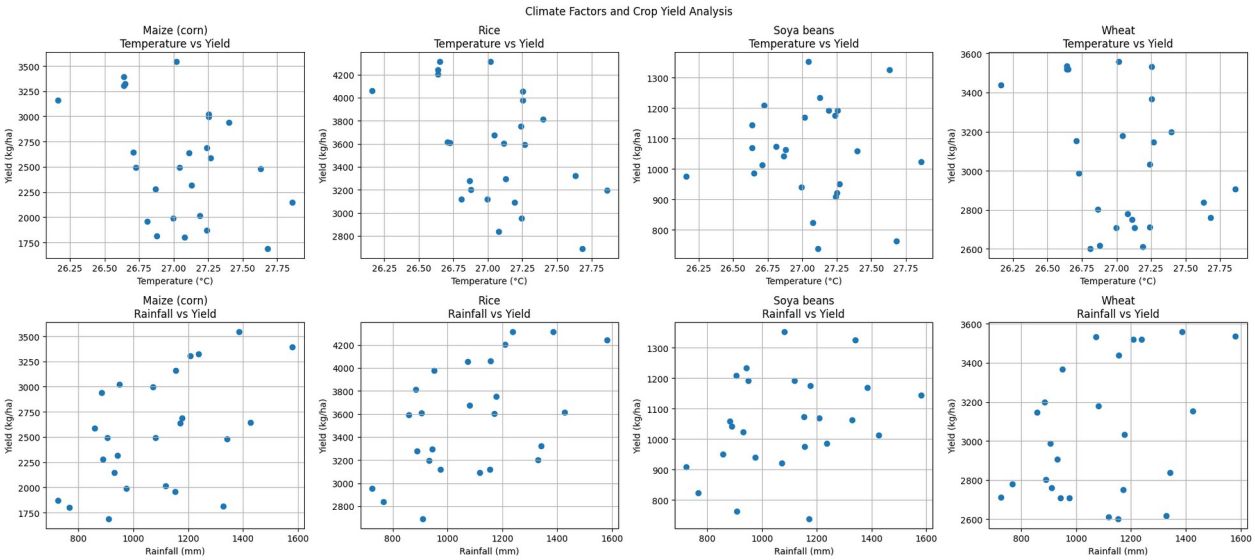


Figure 2: Temperature vs Yield (top row) and Rainfall vs Yield (bottom row) for all four crops

What the graphs show

In the temperature plots, higher temperatures generally go with lower yields, but the pattern is not the same for every crop.

In the rainfall plots, more rain generally goes with higher yields. The points are spread out, so rainfall alone cannot explain all the changes in yield.

I checked these patterns further using correlation.

8. Correlation Analysis

Correlation tells us how strongly two things move together. It goes from -1 to +1.

- A positive value means that when one goes up, the other tends to go up too.
- A negative value means that when one goes up, the other tends to go down.
- A value close to zero means a weak straight-line link.

8.1 Results

Crop	Temperature Correlation	Rainfall Correlation
Maize	-0.399	0.505
Rice	-0.462	0.522
Soybean	-0.038	0.314
Wheat	-0.352	0.382

8.2 What the Results Mean

- Maize: Temperature is negatively linked (-0.399) and rainfall is positively linked (0.505).
- Rice: It has the strongest negative link with temperature (-0.462) and the strongest positive link with rainfall (0.522) of all four crops.
- Soybean: The link with temperature is very weak (-0.038). The link with rainfall is positive (0.314).
- Wheat: Temperature is negatively linked (-0.352) and rainfall is positively linked (0.382).

> **Overall pattern**
>
> All four crops show a negative link with temperature and a positive link with rainfall. These results show a link between the variables. They do not prove that temperature or rainfall alone caused the change in yield.

9. Machine Learning Model

I used machine learning to predict crop yield from past data.

9.1 Algorithm

Multiple Linear Regression is a supervised learning method. It predicts a number using more than one input. I built four separate models, one for each crop.

9.2 Inputs and Output

Role	Variables
Inputs	Year, Average Temperature, Total Rainfall
Output	Crop Yield (kg/ha)

9.3 Training and Testing

I split the data by year. The models learned from the first 20 years and were tested on the last 5 years.

Data	Years
Training data	2000–2019
Testing data	2020–2024

> **Model approach**
> A separate Multiple Linear Regression model was created for each crop using year, average temperature and total rainfall as inputs.

10. Model Results

I used three measures to check how good the models are.

MAE (Mean Absolute Error)

The average size of the mistakes between actual and predicted values. Lower is better.

RMSE (Root Mean Squared Error)

Similar to MAE, but it gives more weight to big mistakes. Lower is better.

R^2 Score

Shows how well the model fits the data. Higher is better, but it must be read with care when the test data is small.

10.1 Overall Results

Measure	Result
MAE	124.56 kg/ha
RMSE	151.12 kg/ha
R^2 Score	0.984

10.2 Results for Each Crop

Crop	MAE (kg/ha)	RMSE (kg/ha)
Maize	135.27	161.11
Rice	54.06	93.94
Soybean	119.66	132.92
Wheat	189.26	197.24

What the results show

- Rice has the lowest MAE (54.06 kg/ha), so its predictions are the closest.
- Wheat has the highest MAE (189.26 kg/ha).
- Maize and Soybean have medium errors.

> **Note on R^2**
>
> The overall R^2 of 0.984 uses all 20 test records together. The four crops have very different yield levels, so this combined score can look better than it really is. That is why the errors for each crop matter too.

11. Actual vs Predicted Yield

I tested the final models on the years 2020 to 2024 and compared the predicted yields with the actual yields.

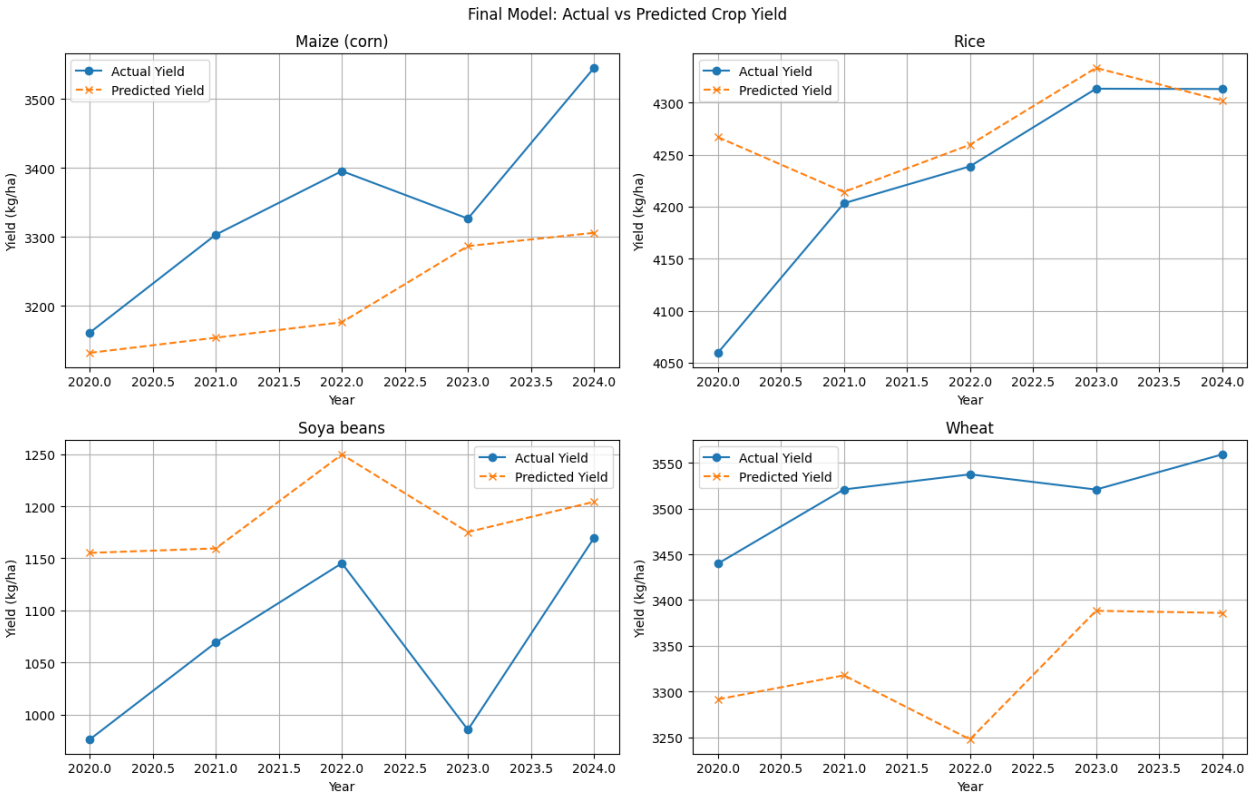


Figure 3: Actual vs predicted yield for each crop, 2020–2024

What the graph shows

- Maize: The model follows the upward direction, but its predictions are lower than the actual values in every year.
- Rice: The predictions are fairly close to the actual values.
- Soybean: The predictions are mostly higher than the actual values.
- Wheat: The predictions follow some of the changes but stay lower than the actual values.

Model quality is different for each crop.

12. Key Findings

1. The final dataset has 100 records for four crops from 2000 to 2024.
2. There are no missing values or duplicate rows.
3. Temperature has a negative link with the yield of all four crops.
4. Rainfall has a positive link with the yield of all four crops.
5. Rice has the strongest positive link with rainfall.
6. Soybean has a very weak link with temperature.
7. The overall model got an R^2 score of 0.984.
8. Rice had the lowest prediction error, and Wheat had the highest.

13. What This Means for Food and Farming

Crop yield is a key part of how much food is available. Temperature and rainfall may affect yield, so studying them can help us understand climate risks for farming.

The findings of this project help us understand past yield patterns and show where more research is needed.

Food supply also depends on the area of land farmed, irrigation, storage, transport and crop losses. This project studies yield only. It does not calculate total food supply.

14. Limitations

1. The climate data is from one central India location, not the whole country.
2. The yield data is for all of India. It does not show differences between states.
3. Only four crops were studied.
4. There are only 100 records, with just five test years for each crop.
5. Soil quality, irrigation, fertiliser use and farming methods were not included.
6. A link between two things does not prove that one causes the other.
7. The overall R^2 score combines four crops with different yield levels, so it should be read with care.

15. Future Improvements

1. Use climate data from many farming regions of India.
2. Use state-wise crop-yield data.
3. Add more variables such as humidity, soil quality and irrigation.
4. Study extreme weather such as droughts and floods.
5. Try and compare other machine learning models.
6. Build an interactive dashboard for the analysis.
7. Add total crop production and food-price data to understand food supply better.

16. Conclusion

This project studied how temperature and rainfall are linked to the yield of four major crops in India: Maize, Rice, Soybean and Wheat.

I collected data from 2000 to 2024 and processed it with Python. All four crops showed a negative link with temperature and a positive link with rainfall.

I built a separate Multiple Linear Regression model for each crop. The overall model got an R^2 score of 0.984, an MAE of 124.56 kg/ha and an RMSE of 151.12 kg/ha.

The project shows how data science and machine learning can be used to study farming data and estimate crop yield. It also shows that better regional climate data and more farming details are needed to understand the effect of climate change more accurately.

17. References

1. FAOSTAT – Food and Agriculture Organization of the United Nations, Crops and Livestock Products Database. https://www.fao.org/faostat/
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